Wine Quality Prediction and Statistical Analysis

Statistics, Regression, Classification and Dashboard using Python Libraries

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Dataset:- Wine Quality (white wine)

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1. Business problem and objectives

Awine-productioncompanywants tounderstandhowthe chemical properties of a wine relate to its quality. The company has historical laboratory measurements and wants a data-driven way to find patterns, predict quality, judge how reliable the predictions are, and share the findings in an interactive dashboard.

Objectives

-Understand and prepare the data (structure, missing values, duplicates, outliers).

-Explore the data visually and find the chemical properties most associated with quality.

-Test the observed relationships with formal statistical hypothesis tests (scipy.stats).

-Build an interpretable OLS regression model (statsmodels) and check its assumptions.

-Predict the numerical quality score with three regression models.

-Classify wines as Good (quality >= 7) or Not Good (quality < 7) with three classification models.

-Compare the models, identify the most useful variables, and recommend a model.

-Present everythingin aninteractive dashboard (Streamlit and Plotly).

2. Dataset and source

Thedata comefrom the **UCIMachine Learning Repository, Wine Quality dataset** (white Vinho Verde wines from northern Portugal). The file winequality-white.csv has **4,898 records**, **11 input variables** (chemical measurements from laboratory tests) and one target, **quality**, a sensory score given by wine tasters. All 12 columns are numeric; there are no categorical columns.

Variable	Role	Unit	Meaning	Min	Max
fixed acidity	Input	g/dm3 (tartaric acid)	Non-volatile acids (mainly tartaric) that do not evaporate easily; give the wine its tartness.	3.8	14.2
volatile acidity	Input	g/dm3 (acetic acid)	Acetic acid level; too much gives an unpleasant vinegar taste.	0.08	1.1
citric acid	Input	g/dm3	Citric acid; adds freshness and flavour in small amounts.	0	1.66
residual sugar	Input	g/dm3	Sugar remaining after fermentation stops; higher values mean a sweeter wine.	0.6	65.8
chlorides	Input	g/dm3 (sodium chloride)	Amount of salt in the wine.	0.009	0.346
free sulfur dioxide	Input	mg/dm3	SO2 not bound to other molecules; prevents microbial growth and oxidation.	2	289
total sulfur dioxide	Input	mg/dm3	Free plus bound SO2; high levels can be noticed in smell and taste.	9	440
density	Input	g/cm3	Density of the wine; depends mainly on alcohol and sugar content.	0.98711	1.03898
pH	Input	pH scale	Acidity/alkalinity; lower pH means a more acidic wine.	2.72	3.82
sulphates	Input	g/dm3 (potassium sulphate)	Wine additive that acts as an antimicrobial and antioxidant.	0.22	1.08
alcohol	Input	% by volume	Alcohol content of the wine.	8	14.2
quality	Target	score (0-10)	Sensory quality score given by wine tasters (median of at least 3 evaluations).	3	9

Suitability: the problem suits both **regression** (predict the score) and **classification** (Good / Not Good). Quality is an integer rating from 3 to 9 that is concentrated around 5 to 7.

3. Data cleaning

Thefollowingchecks were performed on the raw data (notebook 02).

-**Missing values:** none.

-**Duplicate rows:** 937 exact duplicates (19.1% of rows). They were **removed**: if a repeated wine falls in both the training and the testing set, the model is tested on data it has already seen and the scores look better than they really are.

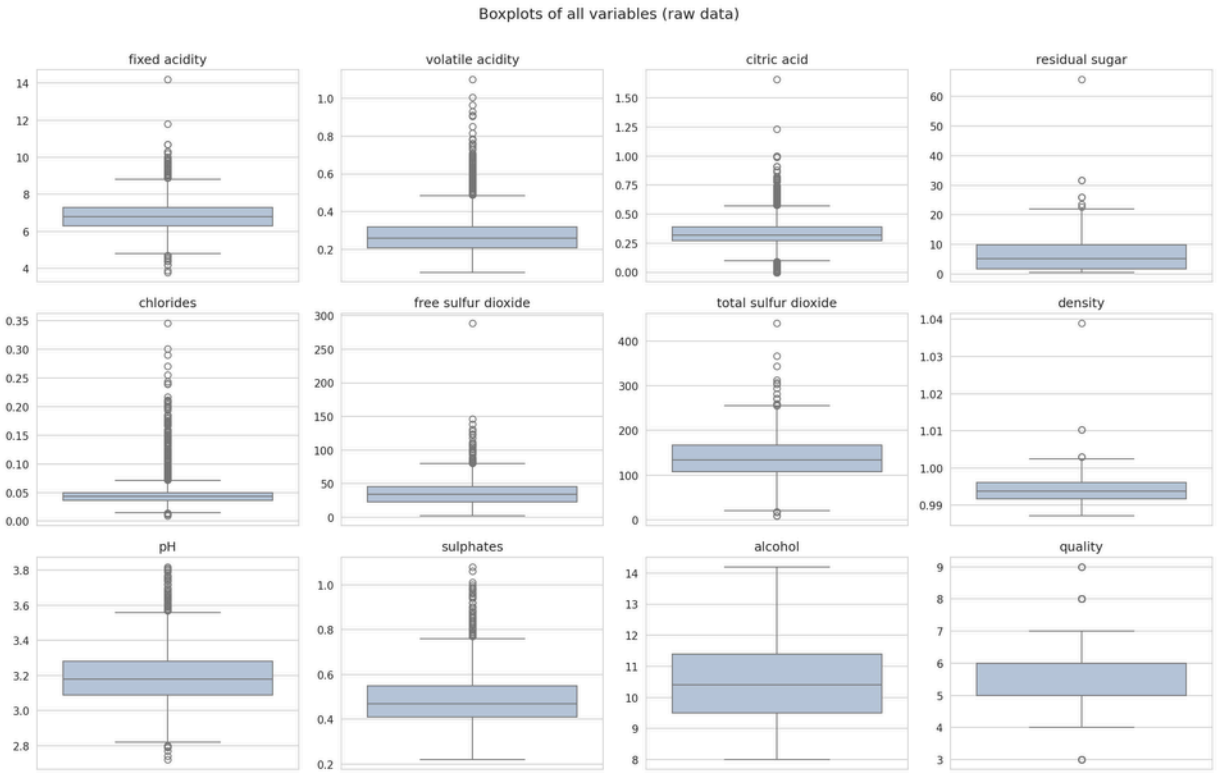
-**Invalid values:** none (no negative values, pH within 0-14, free SO never above total SO, quality within 0-10). Citric acid equals 0 for 18 wines, which is chemically possible, so they were kept.

-**Outliers:** inspected with boxplots and the 1.5 x IQR rule. They are real, plausible measurements (for example very sweet wines), so they were **kept**. Tree-based models are not sensitive to them.

-**Data types:** 11 float64 columns and quality as int64 - all suitable.

Metric	Value
Original row count	4,898
Duplicate rows originally	937
Duplicate rows removed	937
Row count after cleaning	3,961
Duplicate rows remaining	0
Missing values originally	0
Missing values remaining	0
Columns	12
Negative numeric values remaining	0
Outlier values flagged by IQR rule (kept)	1,063

The cleaned data (3,961 rows) were saved as wine_quality_cleaned.csv.



4. Exploratory data analysis

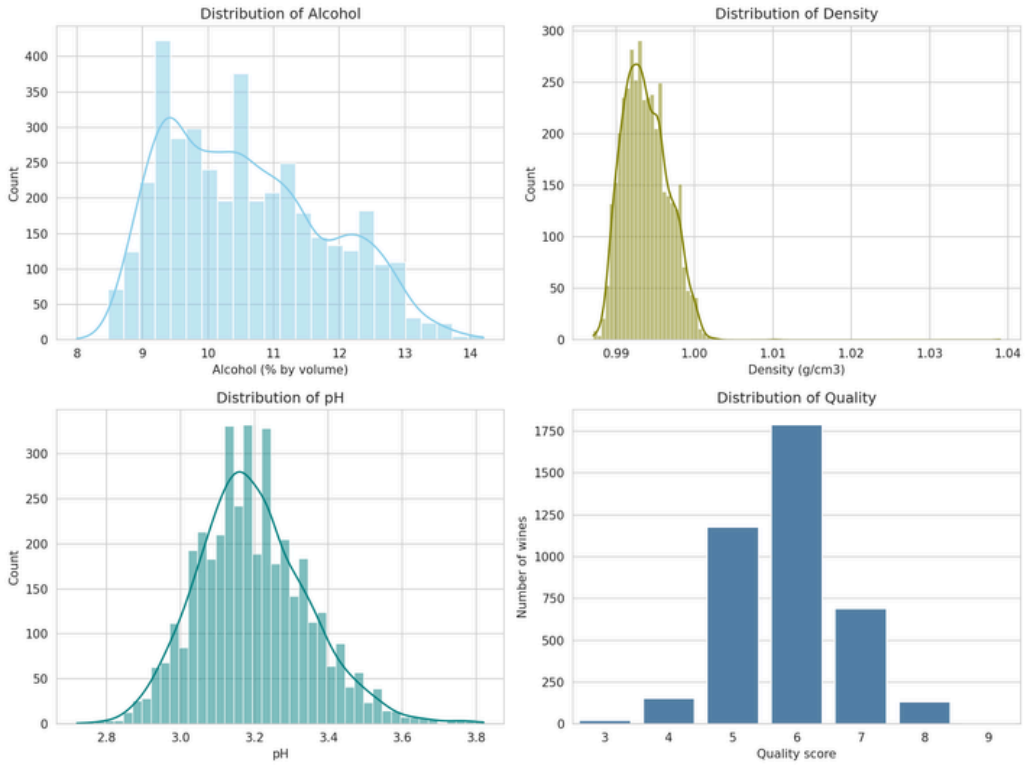
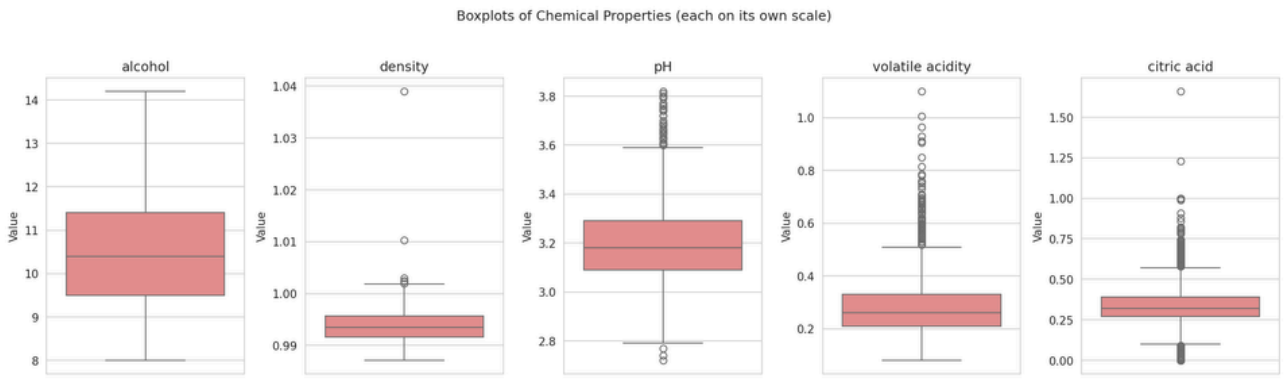
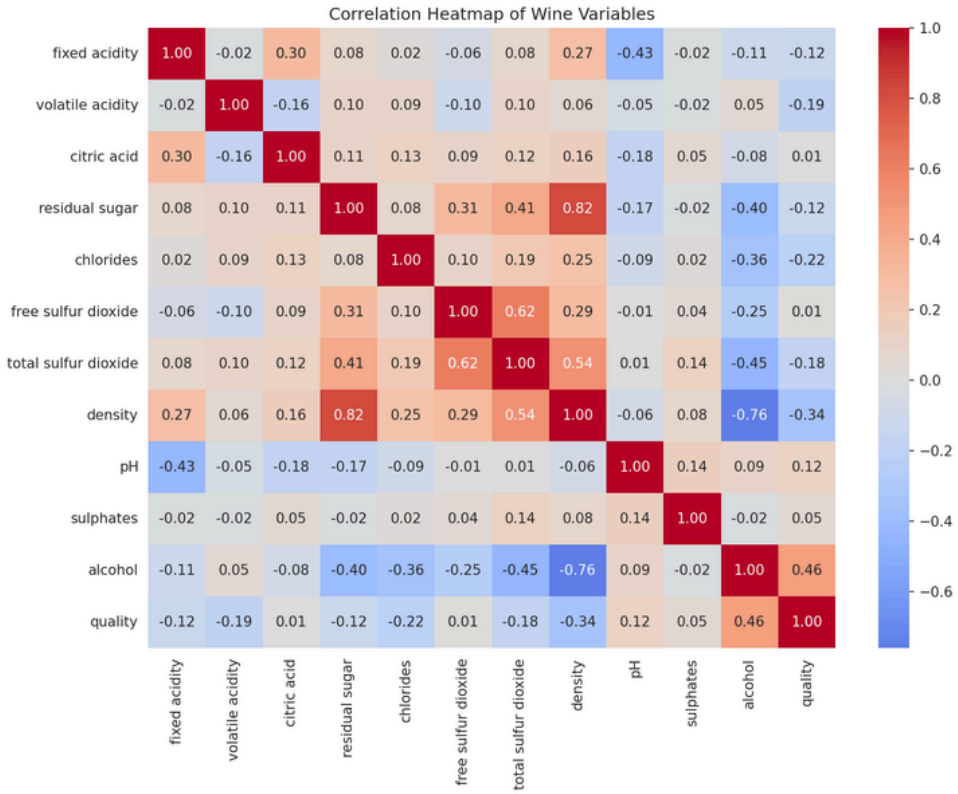
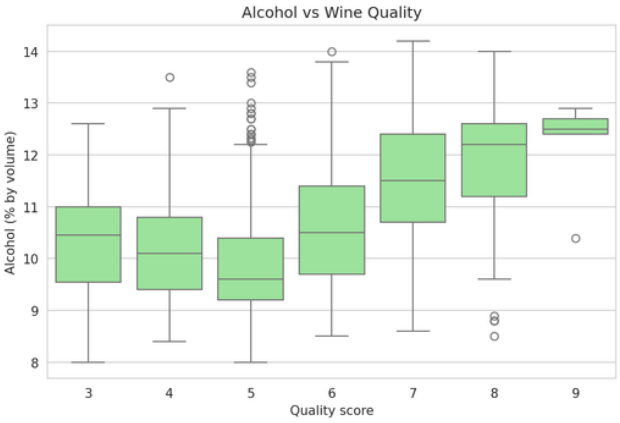


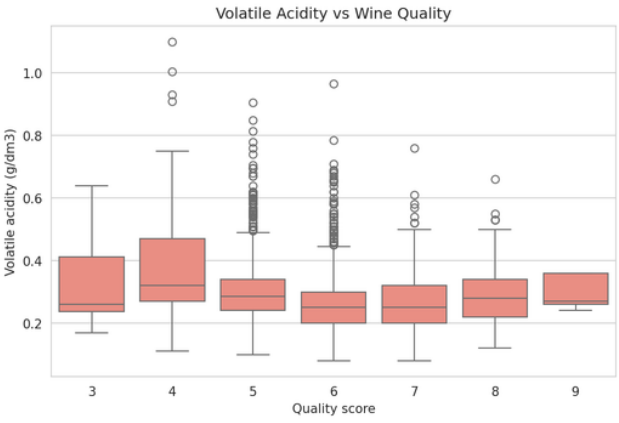
Figure 2. Distributions of alcohol, density, pH and quality.

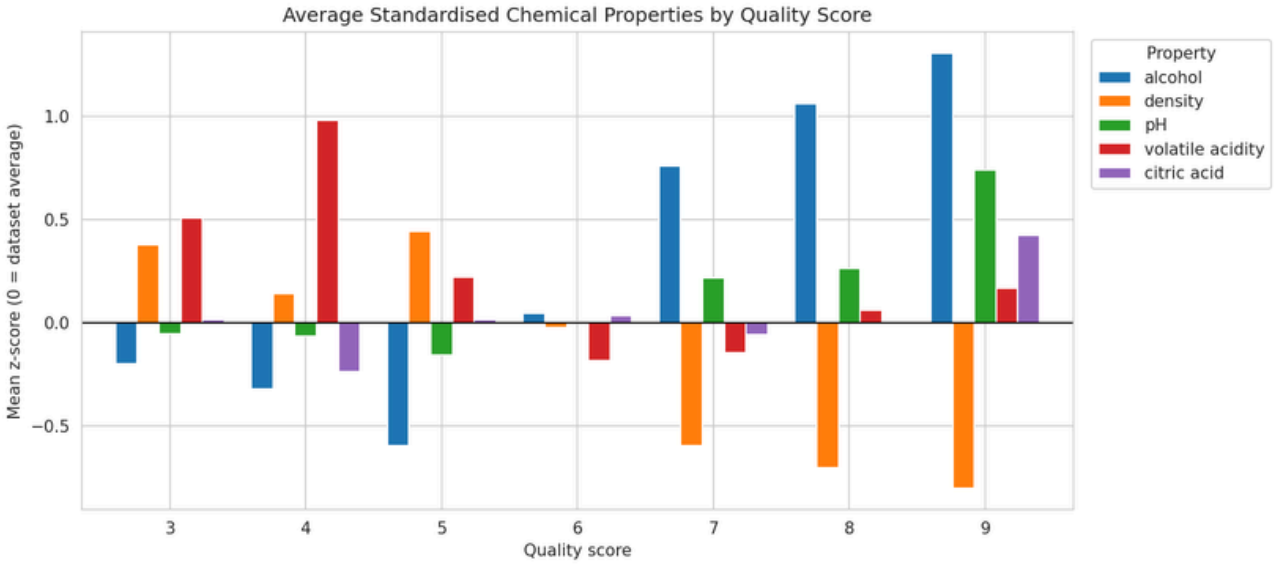
About 92% of wines score 5, 6 or 7 (29.7%, 45.1% and 17.4%); scores 3 and 9 are very rare (0.5% and 0.1%).

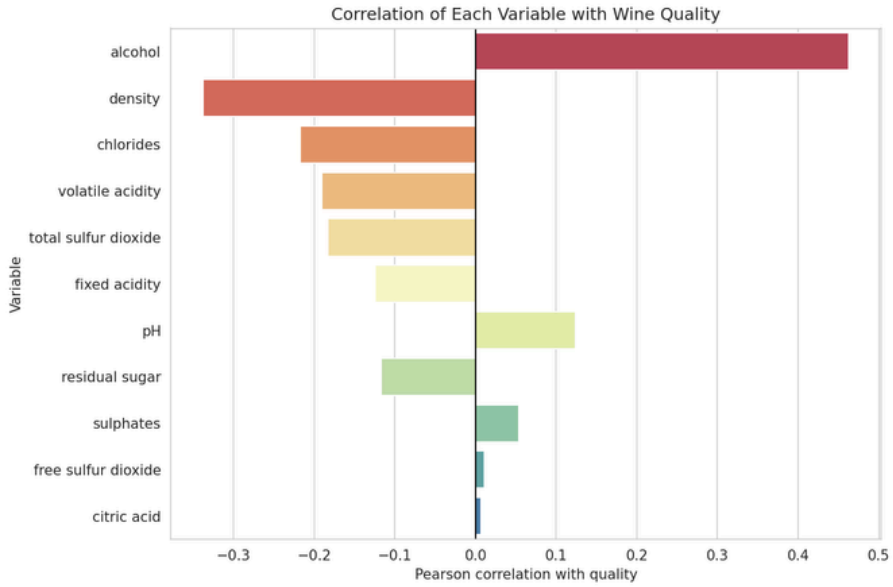



Correlation Heatmap of Wine Variables
fixed acidity 1.00 -0.02 0.30 0.08 0.02 -0.06 0.08 0.27 -0.43 -0.02 -0.11 -0.12
volatile acidity -0.02 1.00 -0.16 0.10 0.09 -0.10 0.10 0.06 -0.05 -0.02 0.05 -0.19
citric acid 0.30 -0.16 1.00 0.11 0.13 0.09 0.12 0.16 -0.18 0.05 -0.08 0.01
residual sugar 0.08 0.10 0.11 1.00 0.08 0.31 0.41 0.82 -0.17 -0.02 -0.40 -0.12
chlorides 0.02 0.09 0.13 0.08 1.00 0.10 0.19 0.25 -0.09 0.02 -0.36 -0.22
free sulfur dioxide -0.06 -0.10 0.09 0.31 0.10 1.00 0.62 0.29 -0.01 0.04 -0.25 0.01
total sulfur dioxide 0.08 0.10 0.12 0.41 0.19 0.62 1.00 0.54 0.01 0.14 -0.45 -0.18
density 0.27 0.06 0.16 0.82 0.25 0.29 0.54 1.00 -0.06 0.08 -0.76 -0.34
pH -0.43 -0.05 -0.18 -0.17 -0.09 -0.01 0.01 -0.06 1.00 0.14 0.09 0.12
sulphates -0.02 -0.02 0.05 -0.02 0.02 0.04 0.14 0.08 0.14 1.00 -0.02 0.05
alcohol -0.11 0.05 -0.08 -0.40 -0.36 -0.25 -0.45 -0.76 0.09 -0.02 1.00 0.46
quality -0.12 -0.19 0.01 -0.12 -0.22 0.01 -0.18 -0.34 0.12 0.05 0.46 1.00
fixed acidity
volatile acidity
citric acid
residual sugar
chlorides
free sulfur dioxide
total sulfur dioxide
density
pH
sulphates
alcohol
quality
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6


Alcohol vs Wine Quality
Alcohol (% by volume)
14
13
12
11
10
9
8
3
4
5
6
7
8
9
Quality score


Volatile Acidity vs Wine Quality
Volatile acidity (g/dm3)
1.0
0.8
0.6
0.4
0.2
3
4
5
6
7
8
9
Quality score





Key observations

-**Alcohol**has thestrongest association with quality (r = +0.46). Median alcohol rises from 9.6% at score 5 to 11.5% at score 7 and 12.5% at score 9; Good wines average 11.58% against 10.33% for the rest.

- **Density** is negatively associated with quality (r = -0.34), but it is strongly tied to alcohol (r = -0.76) and residual sugar (r = +0.82), which warns of multicollinearity.

- **Volatile acidity** (r = -0.19), **chlorides** (r = -0.22) and **total sulfur dioxide** (r = -0.18) are weaker negative associations.

-**Citric acid, free sulfur dioxide and sulphates** show almost no linear relationship with quality.

-Even alcohol explains only about 21% of the variation in quality, so only moderate predictive power is expected.

5. Statistical hypothesis tests

Alltestsusethecleaneddataandasignificancelevel of alpha = 0.05 (notebook 03).

Test	H0	H1	Statistic	P-value	Decision
Pearson correlation	No linear relationship between alcohol and quality	A linear relationship exists	0.4629	1.43e-209	Reject H0
Spearman correlation	No monotonic relationship between alcohol and quality	A monotonic relationship exists	0.4757	7.11e-223	Reject H0
Welch t-test	Mean alcohol is equal in Good and Not Good wines	Mean alcohol differs between the groups	-28.1636	1.30e-135	Reject H0
Mann-Whitney U	Alcohol has the same distribution in Good and Not Good wines	The distributions differ	560713.5000	7.31e-139	Reject H0
Shapiro-Wilk	Alcohol is normally distributed	Alcohol is not normally distributed	0.9645	1.23e-09	Reject H0

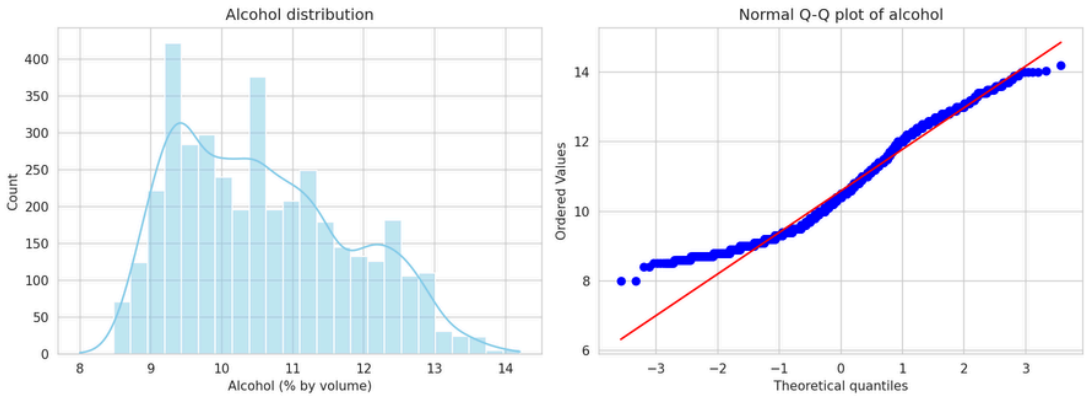
-**Pearson:** r = 0.463 (r^2 = 0.214): a moderate positive linear relationship between alcohol and quality.

-**Spearman:** rho = 0.476: the relationship also holds on ranks, so it is not driven by extreme values.

-**Welch t-test:** Good wines have a higher mean alcohol (11.58% vs 10.33%, difference 1.25 points); Cohen's d = 1.13 (a large effect). The Mann-Whitney U test agrees without assuming normality.

-**Shapiro-Wilk:** W = 0.9645, p about 1.2e-9: alcohol is not normally distributed, which is why rank-based tests were added as a check.

-**Confidence interval:** the mean alcohol is 10.589%, 95% CI 10.551% to 10.627% (n = 3,961).



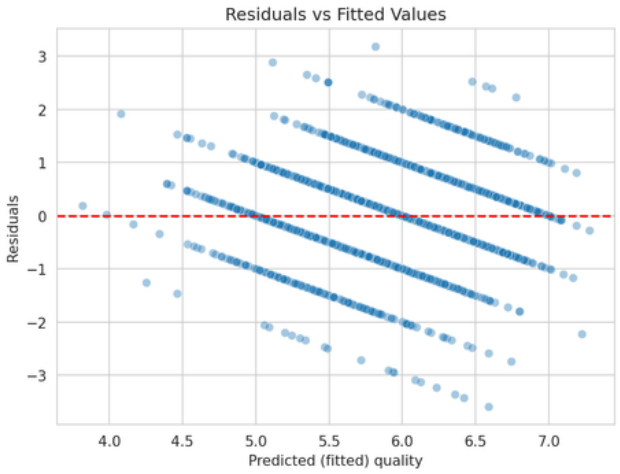
Statistical vs practical significance. With about 4,000 wines even tiny effects give very small p-values, so p-values alone do not show strength. The effect sizes (r = 0.46, d = 1.13) show the alcohol difference is meaningful in practice. But alcohol explains only 21% of the variation in quality, and the tests show association, not causation.

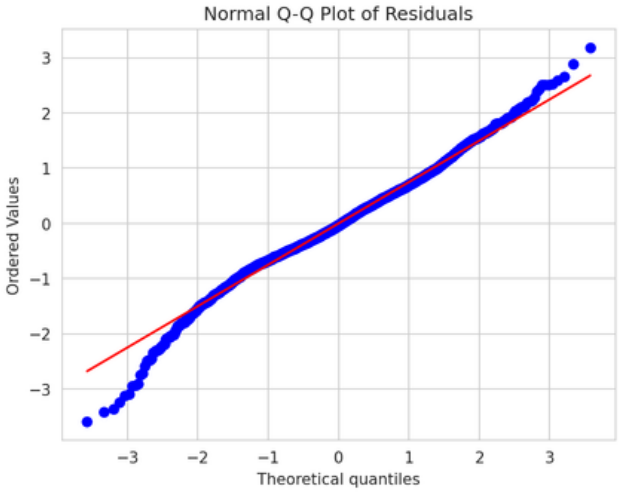
6. OLS regression model

AnOrdinaryLeastSquares model (statsmodels, with an intercept) predicts quality from six variables: alcohol, density, pH, volatile acidity, citric acid and residual sugar (notebook 04).

Variable	Coefficient	P-value	95% CI lower	95% CI upper
const	104.3354	4.29e-14	77.3482	131.3226
alcohol	0.2478	6.29e-34	0.2081	0.2874
density	-104.0343	5.80e-14	-131.0857	-76.9829
pH	0.7268	3.68e-17	0.5584	0.8952
volatile acidity	-1.7751	2.38e-48	-2.0101	-1.5402
citric acid	0.2979	0.0045	0.0925	0.5032
residual sugar	0.0609	1.48e-25	0.0495	0.0722

- **Fit:** $R^2 = 0.2872$, adjusted $R^2 = 0.2861$. The model explains about 29% of the variation in quality. The F-statistic is 265.56 with p = 3.1e-286, so the model as a whole is significant.
- **Coefficients** (other variables held constant): each extra 1% of alcohol goes with about +0.25 quality points; each +0.1 g/dm^3 of volatile acidity with about -0.18; a higher pH with higher quality. All six p-values are below 0.05.
- **Residual checks:** the Breusch-Pagan test finds unequal variance (p = 3.8e-11) and the Jarque-Bera test finds heavier tails than a normal curve (p = 2.2e-43). Durbin-Watson = 1.77. The coefficients are reasonable with this sample size, but the p-values should be read with care.





Multicollinearity. The variance inflation factor (VIF) is above 10 for density (11.23) and above 5 for residual sugar (5.53); the others are below 4.3. The model was therefore refitted **without density** (Model 2): R^2 falls only from 0.2872 to 0.2770, all VIFs drop below 1.25, the alcohol coefficient rises to 0.377 and citric acid becomes non-significant (p = 0.33). Model 2 is the better model for interpreting the individual effects.

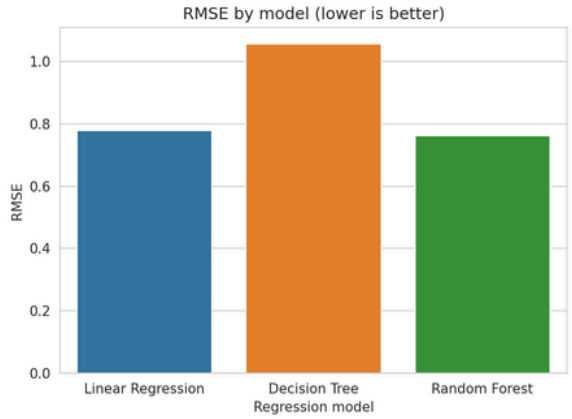
Variable	Model 1 coef	Model 1 p	Model 2 coef (no density)	Model 2 p
const	104.3354	4.29e-14	0.5721	0.0553
alcohol	0.2478	6.29e-34	0.3767	4.09e-231
density	-104.0343	5.80e-14	-	-
pH	0.7268	3.68e-17	0.5195	2.58e-10
volatile acidity	-1.7751	2.38e-48	-1.8983	4.46e-55
citric acid	0.2979	0.0045	0.0991	0.3318
residual sugar	0.0609	1.48e-25	0.0224	5.18e-16

Limitations of the OLS model: it shows association, not causation; quality is an ordinal integer score but OLS treats it as continuous; unequal variance and heavy tails affect the p-values; only 29% of the variance is explained.

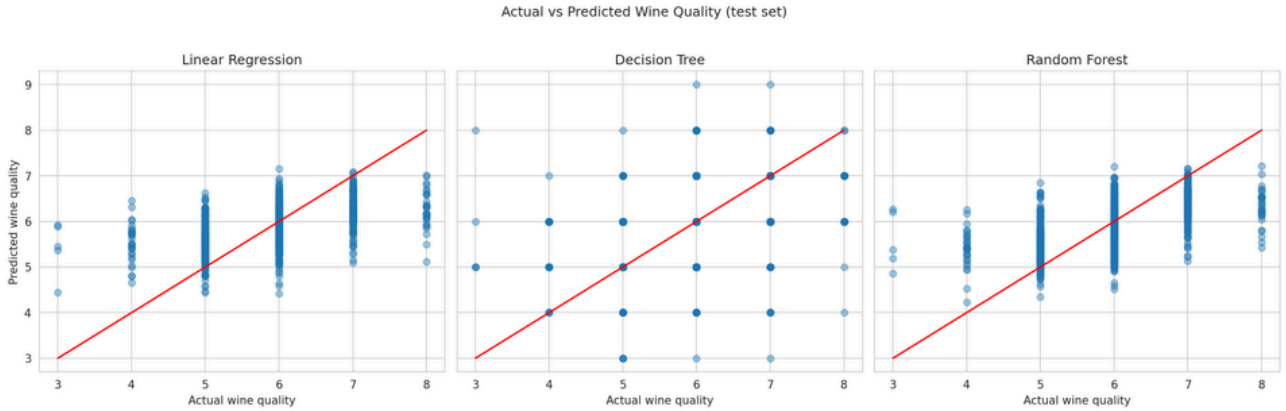
7. Regression models

Quality was predicted from the same six variables as the OLS model, using an 80/20 train/test split with a fixed random seed (42): 3,168 training and 793 testing wines (notebook 05).

Model	MAE	MSE	RMSE	R^2
Linear Regression	0.6012	0.6051	0.7779	0.2621
Decision Tree	0.7478	1.1160	1.0564	-0.3608
Random Forest	0.5833	0.5811	0.7623	0.2914







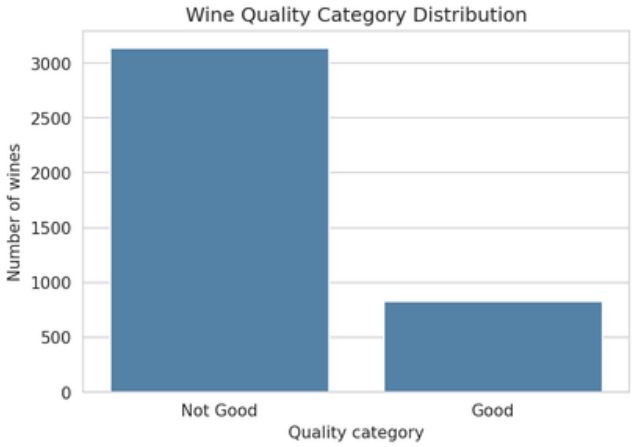
Best model: Random Forest, chosen for the lowest test RMSE (0.762); it also has the lowest MAE and the highest R^2 (0.291). Its margin over Linear Regression is small (RMSE 0.778). The Decision Tree has a

negative R^2 (-0.36): on new data it predicts worse than simply guessing the average quality.

Model	Train R^2	Test R^2	Train RMSE	Test RMSE	Verdict
Linear Regression	0.2932	0.2621	0.7455	0.7779	Stable but too simple (underfits)
Decision Tree	0.9992	-0.3608	0.0251	1.0564	Severe overfitting
Random Forest	0.9034	0.2914	0.2755	0.7623	Overfits, but generalises best

8. Classification models

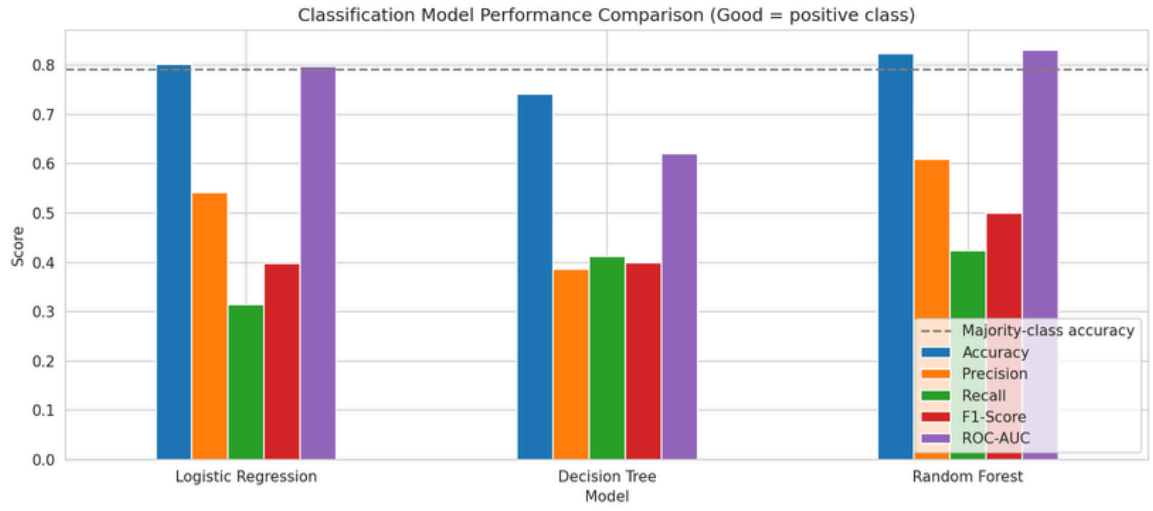
The target quality_category is **Good** if quality >= 7 and **Not Good** otherwise (notebook 06).

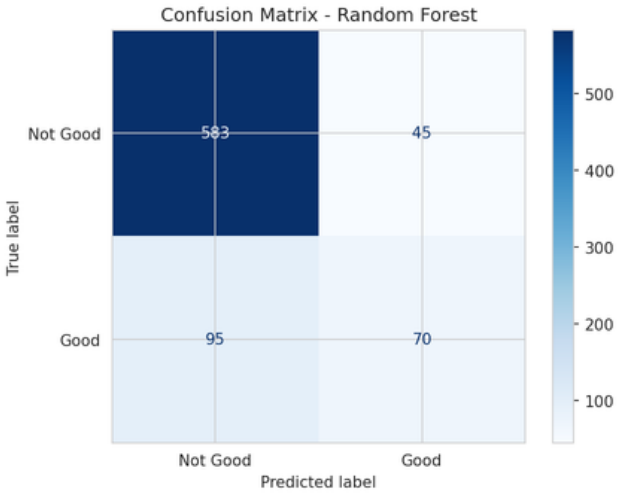


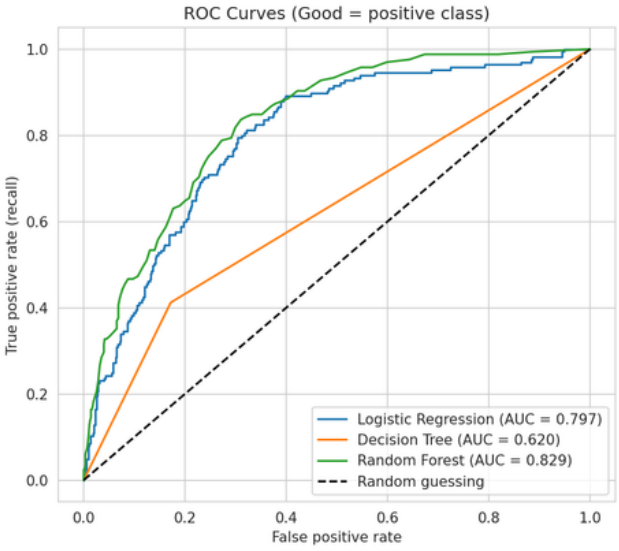
Class	Count	Percentage
Not Good	3,136	79.17%
Good	825	20.83%

The data are imbalanced (about 3.8 to 1). A model that always answers 'Not Good' would be right 79.17% of the time while finding no good wines, so accuracy alone is misleading and precision, recall, F1 and ROC-AUC are used as well. The split was **stratified**, so both sets keep the same class proportions. The original quality column was removed from the predictors, because the label is derived from it and keeping it would cause **target leakage**.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.8020	0.5417	0.3152	0.3985	0.7972
Decision Tree	0.7415	0.3864	0.4121	0.3988	0.6201
Random Forest	0.8235	0.6087	0.4242	0.5000	0.8295







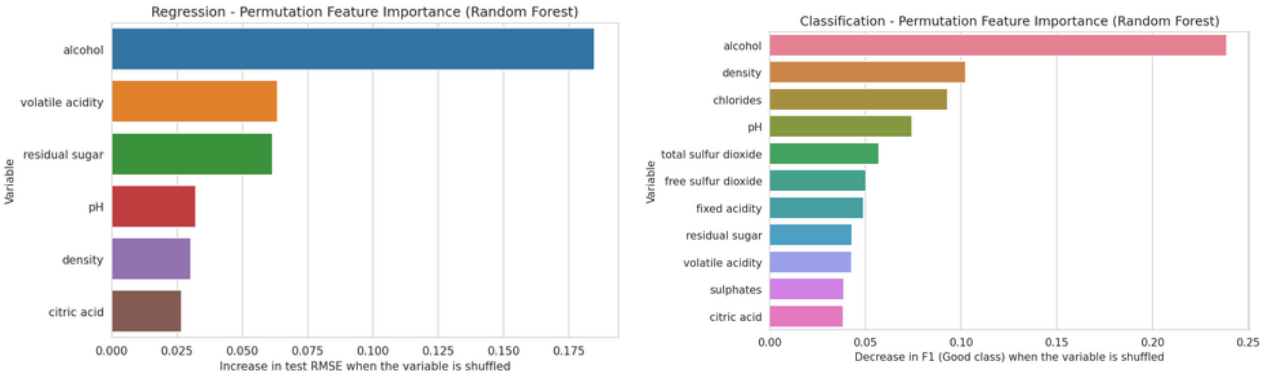
Selected model: Random Forest (highest F1 for the Good class, 0.50, and highest ROC-AUC, 0.83). On the 793 test wines it correctly identified 583 Not Good wines and 70 Good wines; it raised 45 false alarms and missed 95 Good wines.

Strengths and limitations of the selected classifier

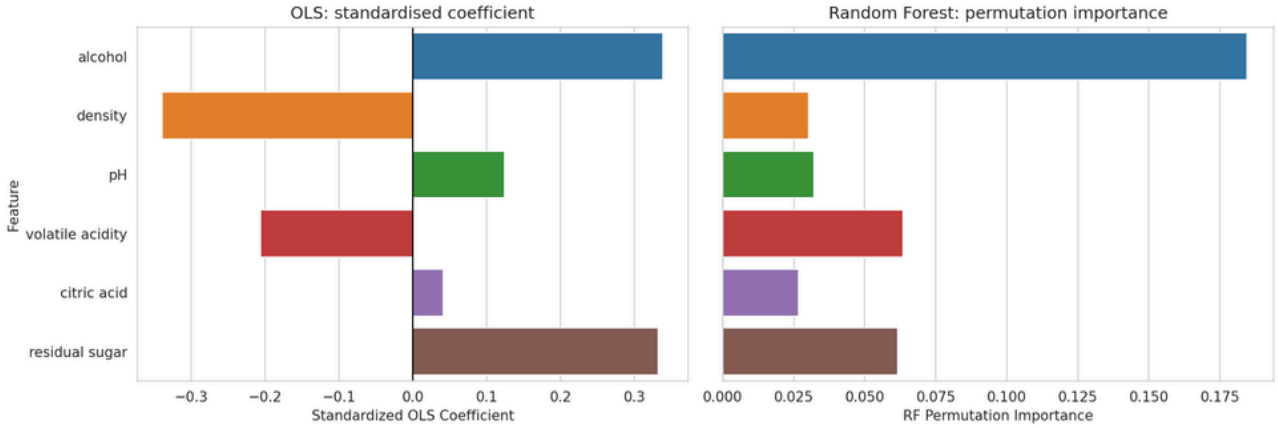
- **Strengths:** best on every metric; separates Good from Not Good clearly better than chance (AUC 0.83); precision of 61%; captures non-linear effects without scaling; gives variable importance.
- **Limitations:** recall for Good wines is only 42%, so it misses most good wines; accuracy (82.4%) is only about 3 points above the 79.2% baseline; it reaches 100% accuracy on the training data (overfitting); it is less interpretable than Logistic Regression; the imbalance was not corrected and a single split was used.

Model	Train accuracy	Test accuracy	Train F1	Test F1	Verdict
Logistic Regression	0.815	0.802	0.431	0.399	Stable, but low recall (underfits)
Decision Tree	1.000	0.742	1.000	0.399	Strong overfitting
Random Forest	1.000	0.824	1.000	0.500	Overfits, but best on test

9. Model interpretation and selection



Most useful variables. Alcohol is clearly the most important variable in both models (about three times the importance of the next variable in regression). Volatile acidity and residual sugar follow in regression; density and chlorides follow in classification. Differences among the weaker variables are small and should not be over-interpreted.



Association vs predictive importance. An OLS coefficient describes the association between a variable and quality while the other variables are held constant, in the sample. Permutation importance measures how much the model's accuracy on unseen data depends on the variable. Density and residual sugar have large OLS coefficients (-0.34 and +0.33) with opposite signs because they are highly collinear, but they add little to prediction (0.030 and 0.062). Alcohol is the dependable finding: large in both views.

Task	Best model	Key result
Regression	Random Forest	RMSE 0.762, R^2 0.291 (Linear Regression close behind: 0.778 and 0.262)
Classification	Random Forest	F1 0.50, ROC-AUC 0.83, accuracy 82.4% (baseline 79.2%)
Explanation of effects	OLS without density	Interpretable and free of multicollinearity

Recommendation: use the **Random Forest** for prediction in both tasks. Use the OLS model without density when the goal is to explain how each chemical property relates to quality. Predictions should be treated as a guide, not as a precise quality score.

10. Interactive dashboard

Thedashboard(app.py)was builtwith **Streamlit and Plotly** (Option B). Start it from the project folder with streamlit run app.py. It trains the same models as the notebooks and has six tabs:

Tab	Content
Overview	Total observations and average quality score, quality distribution, correlation heatmap of the chemical properties. **Interactive filters** (quality range and alcohol range) are in the sidebar.
Regression	Model-performance table and chart; actual-versus-predicted plot with a model selector.
Classification	Accuracy, precision, recall, F1 and ROC-AUC table; confusion matrix with a model selector.
Important variables	Permutation importance for regression and classification.
Findings	Three evidence-based findings computed from the data.
Try a prediction	Sliders for all 11 chemical properties; live predicted quality score and probability of 'Good'.

Dashboard findings

-**Alcohol is the variable most linked to quality** (correlation +0.46); Good wines average 11.58% alcohol versus 10.33%, and alcohol is the top variable in both Random Forest models.

-**Chemistry explains only part of quality:** the best regressor reaches R^2 = 0.29 and RMSE = 0.76 quality points, barely better than Linear Regression (0.26); the unrestricted Decision Tree overfits (-0.36).

-**Good wines are hard to find:** only 20.8% of wines are Good; the Random Forest reaches 82.3% accuracy (always answering Not Good gives 79.2%) with recall of 42% and ROC-AUC 0.83.

Run `streamlit run app.py`

11. Limitations and future improvements

Limitations

-**Weak predictive signal:** the best regression R^2 is about 0.29; chemistry alone does not determine a tasting score.

-**Subjective, ordinal target:** quality is a taster's rating, and OLS and regression models treat it as continuous.

-**Imbalance:** only 21% of wines are Good; extreme scores (3 and 9) have very few examples.

-**Overfitting:** the Decision Tree and Random Forest fit the training data almost perfectly.

-**Narrow data:** white Vinho Verde wines from one region; results may not transfer to other wines.

-**Evaluation:** a single train/test split, no cross-validation and no hyper-parameter tuning.

-**Statistics:** unequal variance and heavy-tailed residuals in OLS; association is not causation.

-**Data cleaning:** removing 937 duplicates reduced the sample from 4,898 to 3,961 wines.

11. Limitations and future improvements

Limitations

-Weak predictive signal: the best regression R2 is about 0.29; chemistry alone does not determine a tasting score.
-**Subjective, ordinal target:** quality is a taster's rating, and OLS and regression models treat it as continuous.

-**Imbalance:** only 21% of wines are Good; extreme scores (3 and 9) have very few examples.

-**Overfitting:** the Decision Tree and Random Forest fit the training data almost perfectly.

-**Narrow data:** white Vinho Verde wines from one region; results may not transfer to other wines.

-**Evaluation:** a single train/test split, no cross-validation and no hyper-parameter tuning.

-**Statistics:** unequal variance and heavy-tailed residuals in OLS; association is not causation.

-**Data cleaning:** removing 937 duplicates reduced the sample from 4,898 to 3,961 wines.

Future improvements

-Use cross-validation andtune the hyper-parameters (tree depth, number of trees).

-Use class weights, threshold tuning or resampling to improve recall for Good wines.
-Try gradient boosting and ordinal or robust regression (for example HC3 standard errors).

Add the red-wine dataset and test whether the findings hold for both wine types.

-Deploy the Streamlit dashboard online and add model explanations for single predictions.

12. Conclusion

The project followed a complete workflow: data understanding, cleaning, exploration, hypothesis testing, OLS modelling, regression, classification, interpretation and a dashboard. **Alcohol** is the strongest and most consistent factor related to wine quality, followed by volatile acidity, density and chlorides. The **Random Forest** is the best predictive model (regression RMSE 0.762, R^2 0.291; classification F1 0.50, ROC-AUC 0.83), but the chemical measurements explain only part of the quality score, so the models are best used as a decision-support aid and not as a replacement for expert tasting.